

Hazem Hassan

Senior AI/ML Engineer — Computer Vision · Conversational AI · Production ML

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PROFESSIONAL SUMMARY

Senior ML Engineer / AI Technical Lead with 9+ years shipping production AI systems end-to-end across three tracks: computer vision and video analytics (multi-object tracking, thermal/IR, edge deployment on NVIDIA Jetson), speech and conversational AI (Arabic ASR at 7.5% WER, XTTS-v2 multilingual TTS at 4.1 average MOS, and a full telephony stack of ASR, diarization, VAD, keyword spotting, intent and NER), and LLM-powered systems (agentic workflows with LangGraph, RAG with LlamaIndex, evaluation and observability with Ragas and LangSmith). Also deep in document AI — 98% F1 layout analysis, 93% F1 graphics understanding, and Arabic OCR in production. First place in two international competitions (MGB-5 2019 dialectal Arabic ASR, ICFHR RASM 2018 manuscript recognition) with three peer-reviewed publications. Comfortable owning the full lifecycle — architecture, code, deployment, evaluation, monitoring — and leading the team that does it.

ROLES AND SPECIALTIES

Roles: Senior ML Engineer, AI Technical Lead, Senior Computer Vision Engineer, LLM/Agentic Systems Engineer, Conversational AI Engineer.

Specialties: Computer Vision Pipelines, OCR, Video Analytics, Multi-Object Tracking, Action Recognition, OCR & Document Understanding, Speech Recognition (ASR), Speech Synthesis (TTS), Speaker Diarization, Agentic Workflows, RAG, LLM Evaluation, Automated Labeling, Human-in-the-Loop Workflows, Active Learning, Weak Supervision, Dataset Quality Assurance, ML APIs, MLOps, Cloud/Edge Deployment.

CORE SKILLS

Programming & Software: Python (expert), C/C++ (intermediate), CUDA / NVIDIA platforms, FastAPI & REST design, multi-threaded real-time systems, Git, Linux.

LLM & Agentic Systems: LangGraph (agent workflows / tool-calling), LlamaIndex (RAG, embedding fine-tuning, retrieval orchestration), Multi-RAG, Embeddings, Reranking, Chunking Strategies, FAISS, Conversation Orchestration, NeMo Guardrails, Escalation Routing, Prompt Engineering.

Speech & NLP: Whisper (fine-tuning, LoRA/full), NVIDIA NeMo (Conformer/CTC, multilingual ASR), Coqui XTTS-v2, Speaker Diarization, VAD, Noise Suppression, Keyword/Hotword Spotting, Telephony/Narrowband (8 kHz) Audio, Intent Classification, Named Entity Recognition, Diacritization, Arabic Dialect Handling (MSA, Egyptian, Saudi, Gulf).

Computer Vision: Object Detection, Segmentation, Multi-Object Tracking, Action Recognition, Anomaly Detection, Face Recognition & Liveness, Optical Flow, Thermal/IR Processing (defogging & dehazing, denoising, motion detection), Diffusion Models.

Document AI & OCR: Layout Analysis, Graphics/Chart Component Analysis & Reasoning, PaddleOCR (Arabic fine-tuning), Text Detection & Recognition, Table & Figure Parsing, Denoising & Post-Correction, K2 Toolkit.

Evaluation & Observability: WER/CER, MOS & naturalness listening tests, Ragas, LangSmith, MT-Bench (LLM-as-judge), AlpacaEval, LM-Eval Harness, Elo Rating, Perplexity, Concurrency/Latency Benchmarking, Error & Failure Analysis.

ML/DL & Data: PyTorch, TensorFlow, Scikit-learn, Model Training & Fine-Tuning, Custom Loss Design, Hyperparameter Tuning, Active Learning, Weak Supervision, SQL / PostgreSQL, Pandas, NumPy, Large-Scale Image/Video/Speech/Text Pipelines.

Deployment & Infrastructure: Model Serving, Docker, Kubernetes, CI/CD, ONNX / TensorRT Optimization, Quantization, NVIDIA Jetson, GCP (Vertex AI), AWS (SageMaker / Ground Truth), Monitoring & Failure Analysis.

Leadership: Technical Leadership, System Design, Cross-functional Collaboration, Mentoring, Code Reviews.

WORK EXPERIENCE

Senior ML Engineer

Beyond AI

March 2024 — Present, Remote, EMEA

LLM & Agentic Systems

- Built agentic LLM workflows with LangGraph (agents as directed graphs with message passing, tool-calling, execution orchestration) and Multi-RAG retrieval on LlamaIndex — hybrid search and reranking lifted Recall@10 ~20% and cut hallucinations ~30%, with time-to-first-token held under 5 s on long documents.
- Stood up the LLM evaluation stack — Ragas + LangSmith for component-level scoring and continuous tracing, plus MT-Bench, AlpacaEval, LM-Eval, Elo, perplexity, and a concurrency benchmark measuring per-token timing under load — making output quality a tracked metric rather than a judgment call.

Speech — ASR & TTS

- Fine-tuned Whisper for Arabic ASR to 7.5% WER on Modern Standard Arabic, then extended the model to Saudi dialect on noisy broadcast audio — reaching accuracy on par with the best published fine-tuned results for that dialect.

- Shipped a multilingual TTS voice stack on Coqui XTTS-v2: two speakers across English, Arabic MSA and Saudi at 4.1 average MOS / 3.5 naturalness, with a single voice identity holding across all three variants.

Computer Vision & Edge

- Built an in-house visual-prompt detector (region-prompt contrastive alignment) that adapts to new domains from a handful of reference images with zero re-training or re-labeling — +3 AP over YOLOE.
- Deployed real-time video analytics on NVIDIA Jetson across 10+ camera feeds / 5+ sites at 25–35 FPS and <150 ms latency; tuned detection + multi-object tracking to 94% flow accuracy with 14% fewer ID switches.
- Shipped PPE-compliance detection for Aramco at 94% precision / 90% recall, cutting false alarms 25% through post-processing and threshold calibration.
- Quantized OpenVLA for task-specific robotic manipulation so the policy runs on-device instead of off-loading to a server; built thermal/IR defogging and optical-flow anomaly detection for low-visibility IR feeds.

Document AI & OCR

- Built Arabic layout and complex-graphics analysis — 98% F1 on layout regions, 93% F1 on graphics components, and 84% on graphics reasoning (chart and diagram semantics, figure-caption linkage).
- Fine-tuned PaddleOCR for Arabic to 14% WER general and ~8% WER domain-specific, handling cursive script, ligatures, and right-to-left layout.

Technical Lead

IDefy (Digital Identity/eKYC)

October 2022 — February 2024, Giza, Egypt

- Led digital identity onboarding products (SaaS/SDK/web/mobile), delivering 3 production deployments end-to-end.
- Built validation and quality-control workflows for OCR, face matching, liveness, and identity-verification datasets, improving verification success rate by 20% through data quality tuning and systematic failure analysis.
- Developed configurable OCR/document-processing pipelines that adapted to new structured documents using limited samples, annotation rules, and validation checks, reaching 96% recognition accuracy.
- Analyzed production failure cases across OCR, face verification, and liveness detection to identify noisy data, poor image quality, edge cases, and model retraining opportunities.

Senior ML Engineer (Computer Vision)

RDI

January 2021 — October 2022, Giza, Egypt

- Led the delivery of the Sotoor Arabic OCR System, achieving 12% WER on Arabic manuscripts.
- Built document OCR pipeline modules (layout analysis, segmentation, font detection, recognition, denoising, correction), reducing OCR error by ~7% on noisy scans.
- Adapted the K2 Toolkit for OCR training/decoding, improving training efficiency by ~25% and gaining ~4% WER, boosting sequence recognition stability.

ML Researcher

RDI

January 2017 — January 2021, Giza, Egypt

- Collaborated on a team building NLP systems for the Arabic language across speech and text.
- Developed and deployed production systems, including Speech Recognition, Optical Character Recognition, Text Diacritization, Voice Conversion, and Quran Recitation Verification.

PROJECTS

Agentic Agent Call Center

- Built an agentic AI call-center assistant using LLM tool-calling and Multi-RAG retrieval to automate intent handling, answer generation, and CRM/ticketing actions.
- Delivered the supporting speech stack: multilingual ASR (NVIDIA NeMo) covering Arabic MSA, Egyptian, Saudi and Gulf plus English on 8 kHz telephony, 4 TTS voices (2 male / 2 female) at 3.2 MOS naturalness, speaker diarization, VAD + noise suppression, keyword/hotword spotting, intent classification, and Named Entity Recognition.
- Designed a reliable conversation orchestration layer with guardrails, escalation-to-human routing, and quality evaluation to ensure safe and consistent customer-support automation.

Fashimi APP — Lead Engineer

- Built an AI try-on platform using LLMs for recommendation + prompt generation and diffusion models for avatar generation and photorealistic try-on rendering.
- Developed a recommendation engine leveraging mood/occasion/climate/user preferences to produce personalized outfit prompts.

Valorant Overlay — Developer

- Designed a real-time multi-threaded system extracting gameplay data from Valorant for esports competition overlays.
- Implemented object detection, pattern recognition, and color analysis, achieving a response latency of 50–72 msec.

EDUCATION

Bachelor of Science in Engineering

Cairo University — Electrical Electronics and Communication Engineering • 2016

PUBLICATIONS

Arabic Multi-Genre Broadcast Speech Recognition — Conference Paper, December 2019 — IEEE ASRU 2019
Prognostic AI for Post-Operative Infections — Conference Paper, November 2023 — FCI Annual Conference
QuranScript Verification System — Conference Paper, November 2016 — Sixteenth ESOLE Conference of Language Engineering

HOBBIES

Swimming	Gaming	Padel
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